BASICS OF GenAI

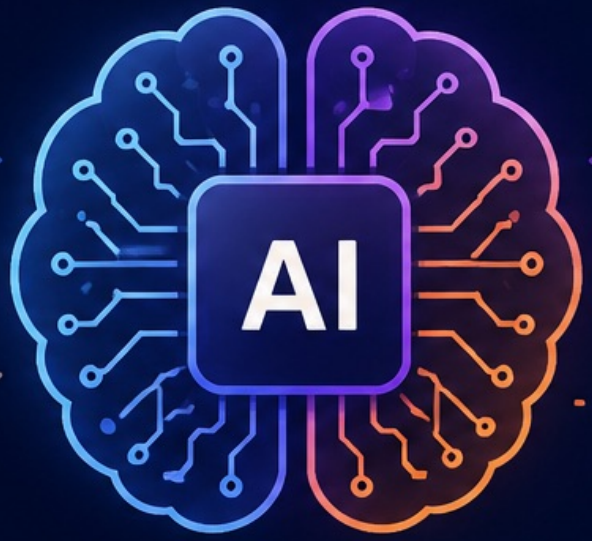
Understand. Create. Innovate.



TEXT GENERATION



IMAGE GENERATION



CODE GENERATION



IDEAS TO IMPACT





UNDERSTAND



CREATE



INNOVATE

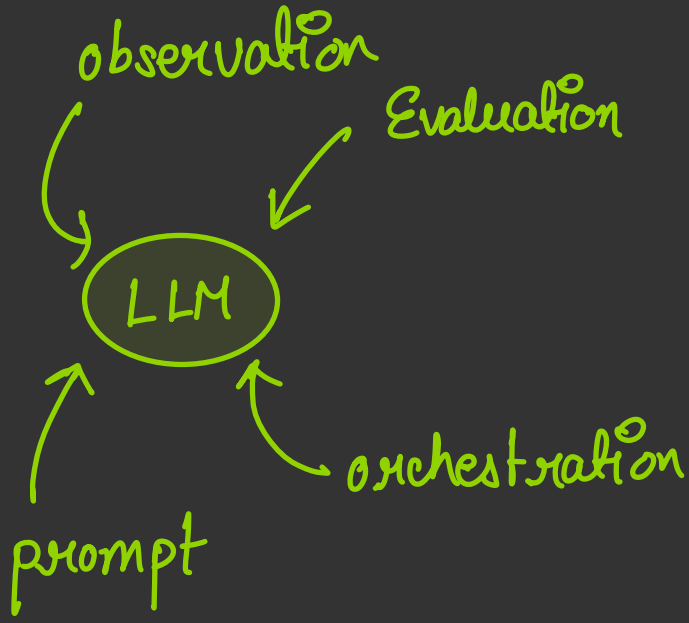


EMPOWER

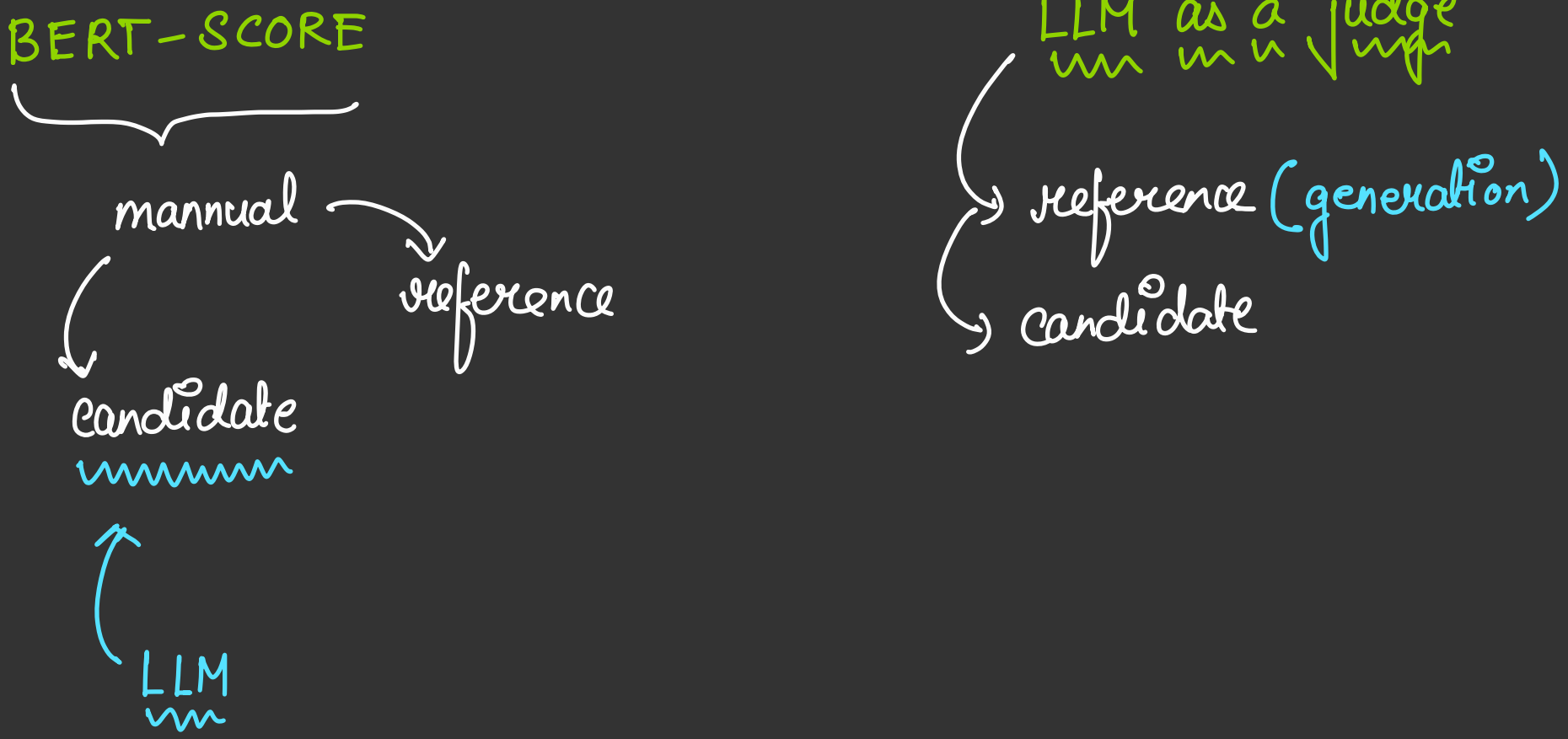
Lecture: Prompt Engineering

Agenda

1. LLM as a judge
2. Other evaluation metrics
3. Opik dashboard
4. Understanding prompt Engineering



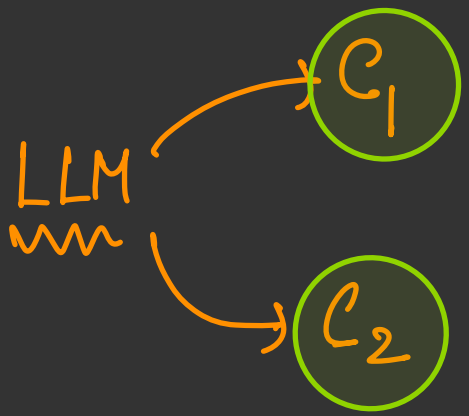
Recap


BERT-SCORE
manual
reference
candidate
LLM
LLM as a judge
reference (generation)
candidate

LLM as a Judge

Reference

generation (style & Tone)



RLHF

Types

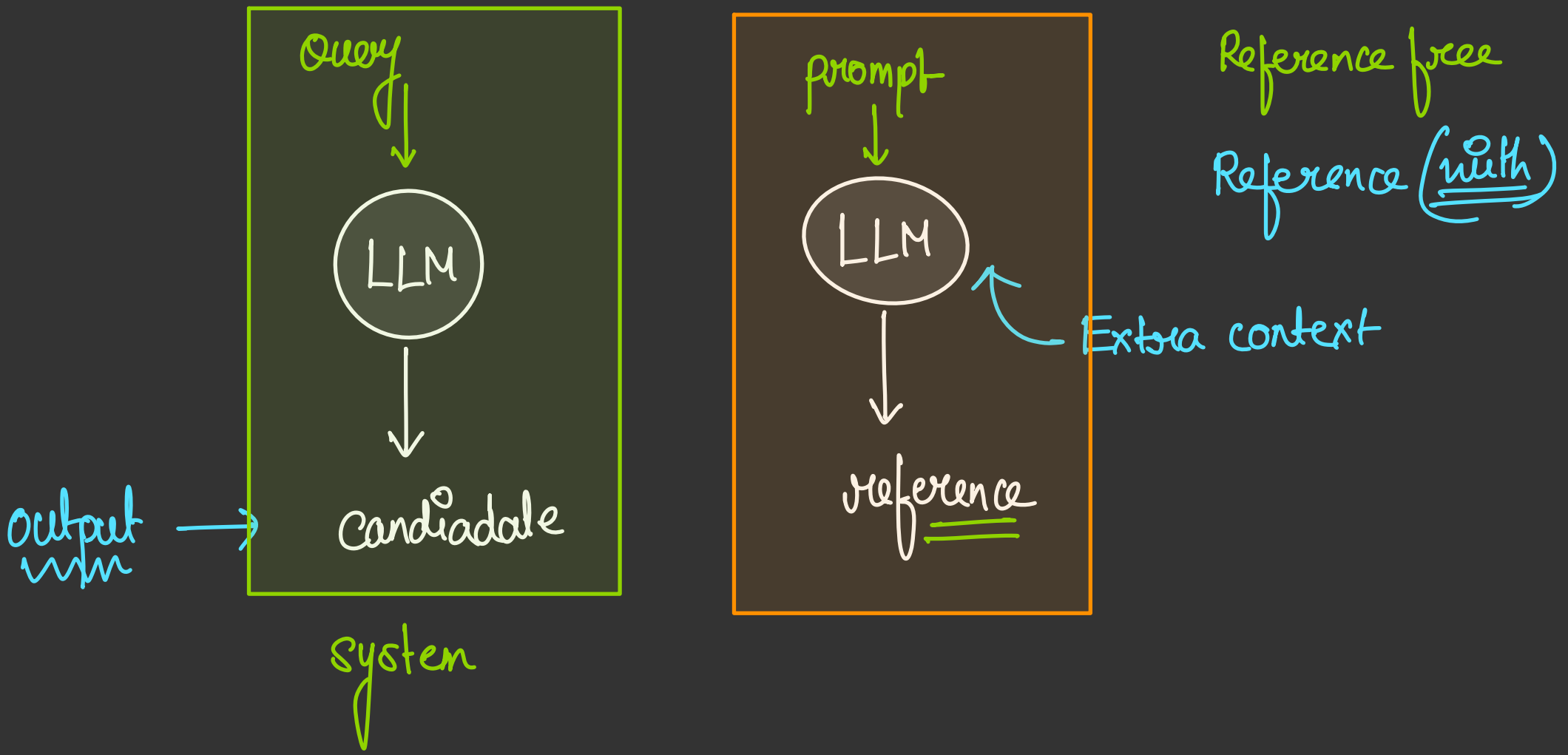
1) pairwise comparision

provide the LLM with two response

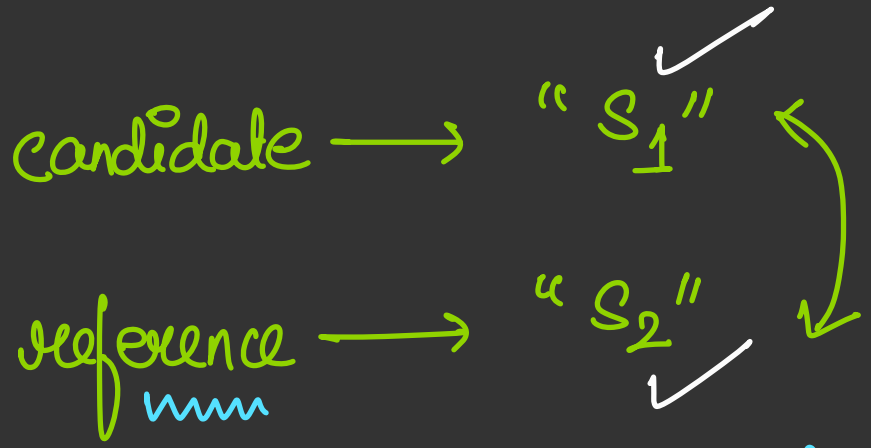
2) Evaluation by criteria (Reference free)

3) Evaluation by criteria (Reference based)

additional context along side the reference.


Query
LLM
Candidate
output
system
prompt
LLM
reference
Extra context
Reference free
Reference (with)

Evaluation by criteria

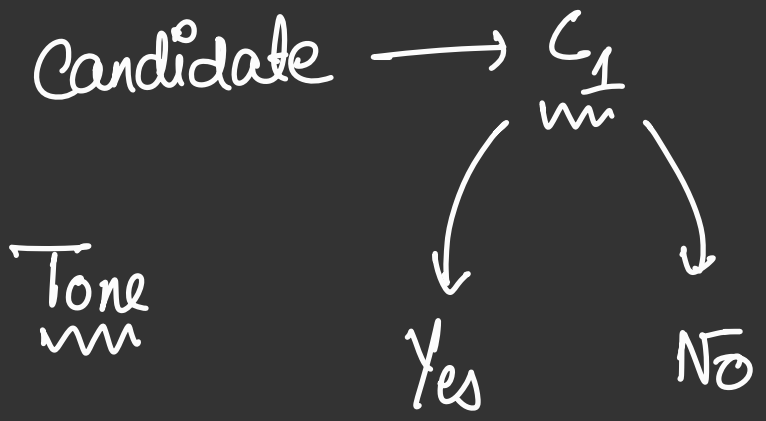


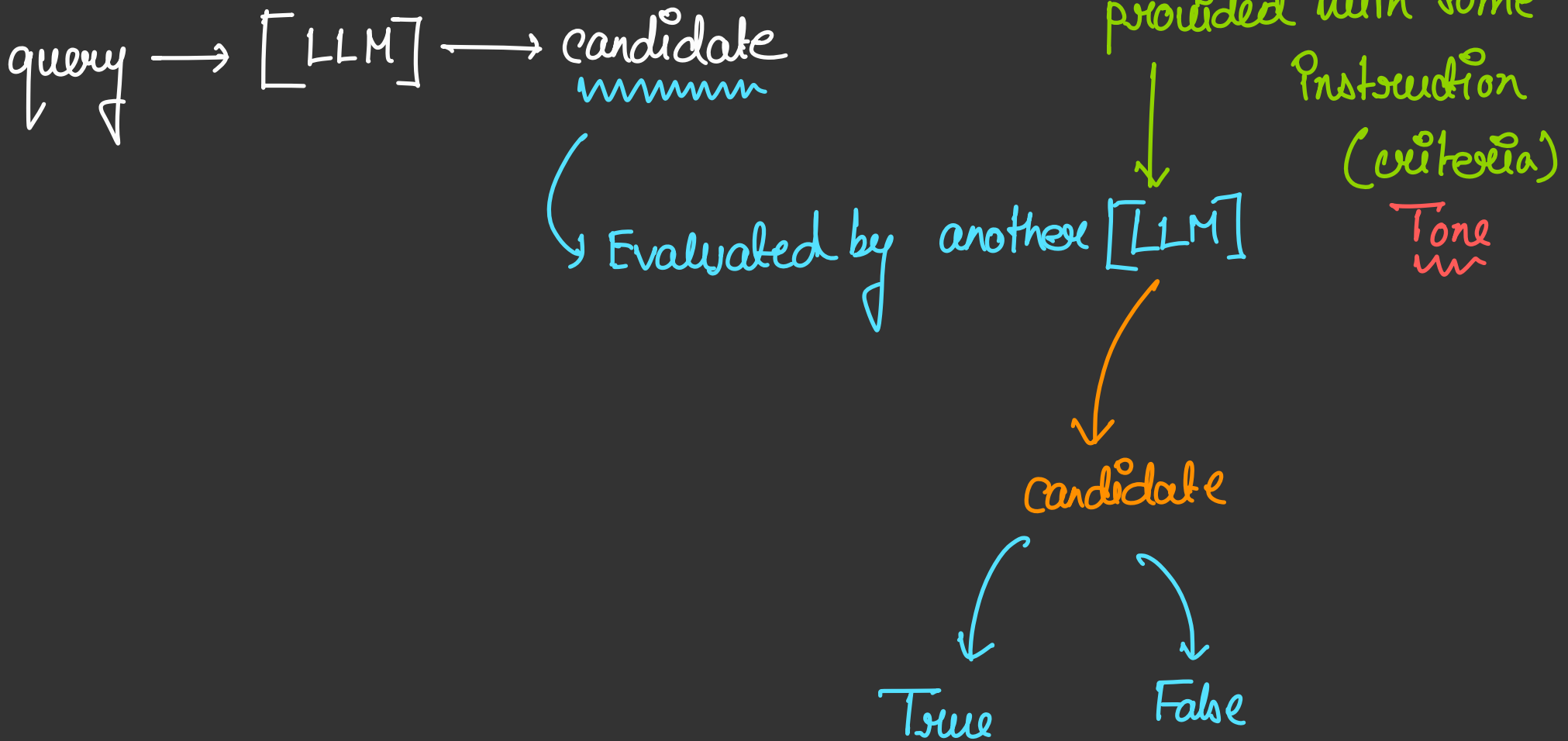
Criteria: Tone: formal

How to create an LLM judge

1) Define the evaluation scenario (Tone, correctness, ...)

Scope: Binary classification




query
[LLM]
candidate
Evaluated by
another
[LLM]
provided with some
Instruction
(criteria)
Tone
candidate
True
False

2) Prepare the Evaluation Dataset

↳ small (diverse samples)

Candidate Vs Reference

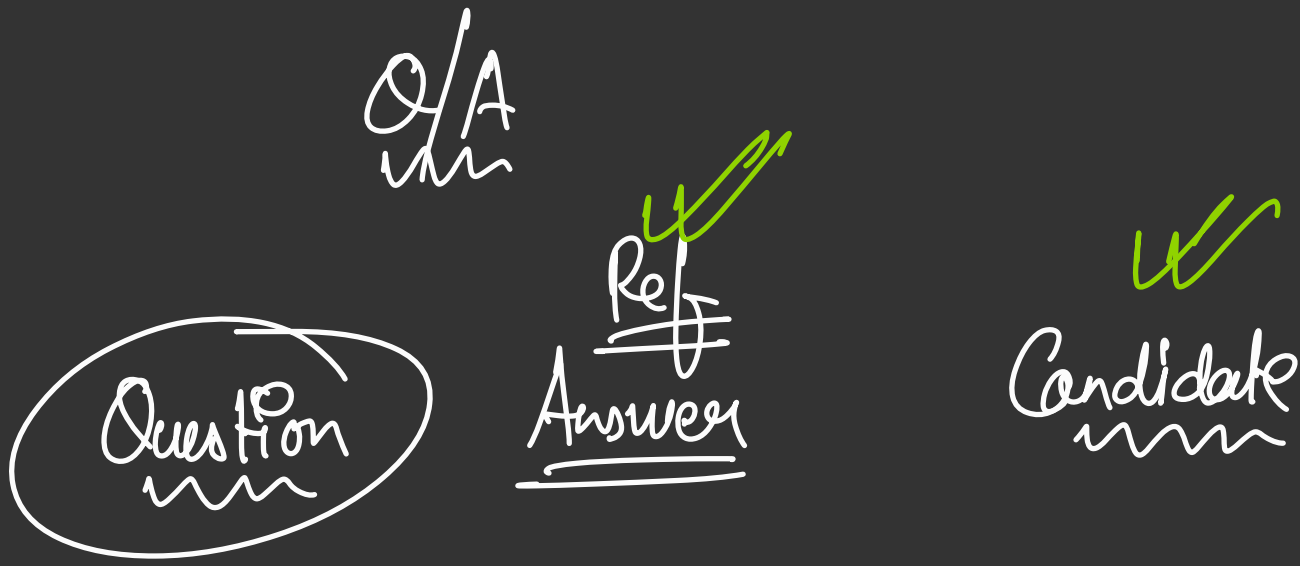
↳ "Some of the students are qualified for maths pre-exam."

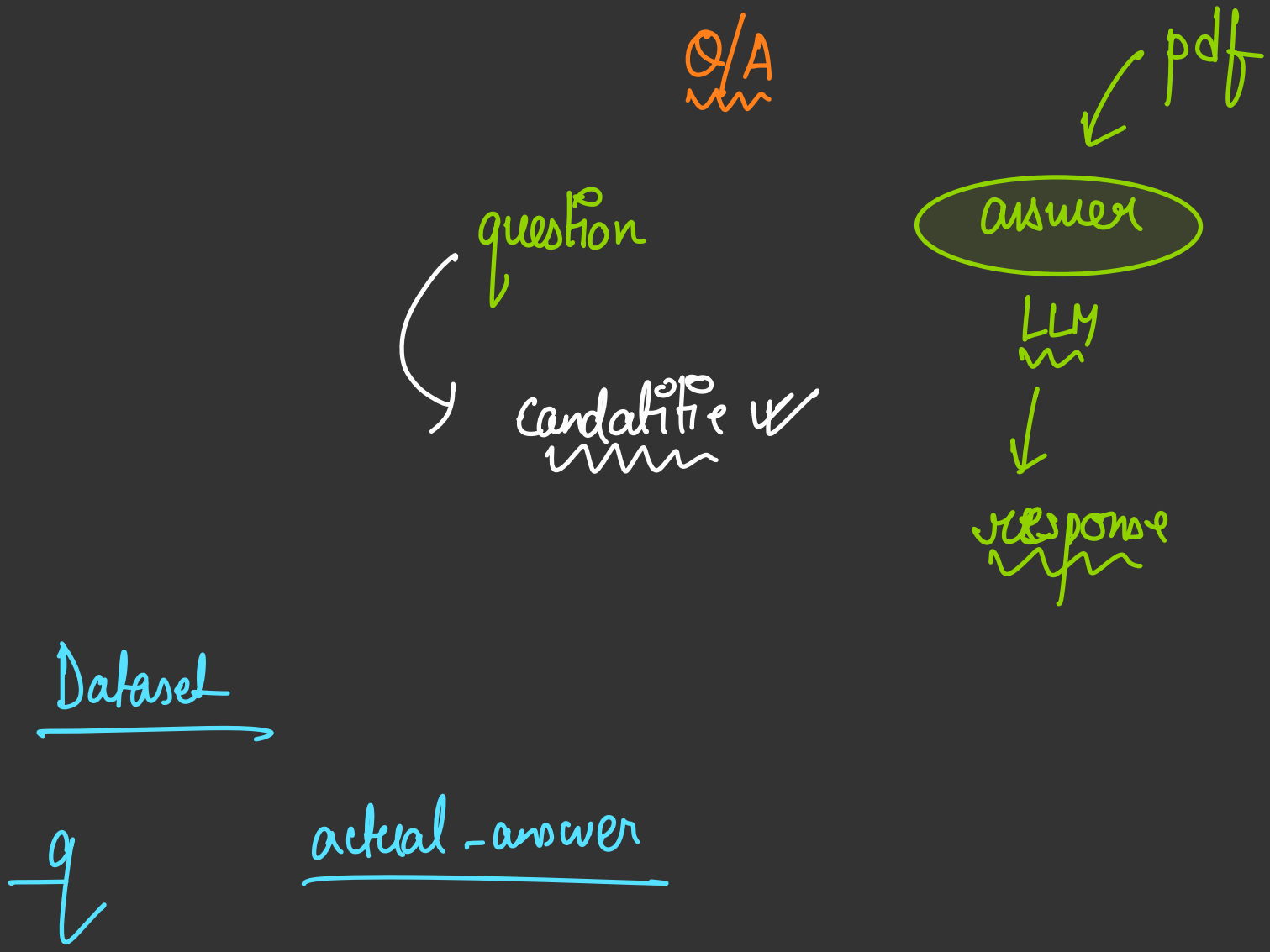
3) Label the dataset

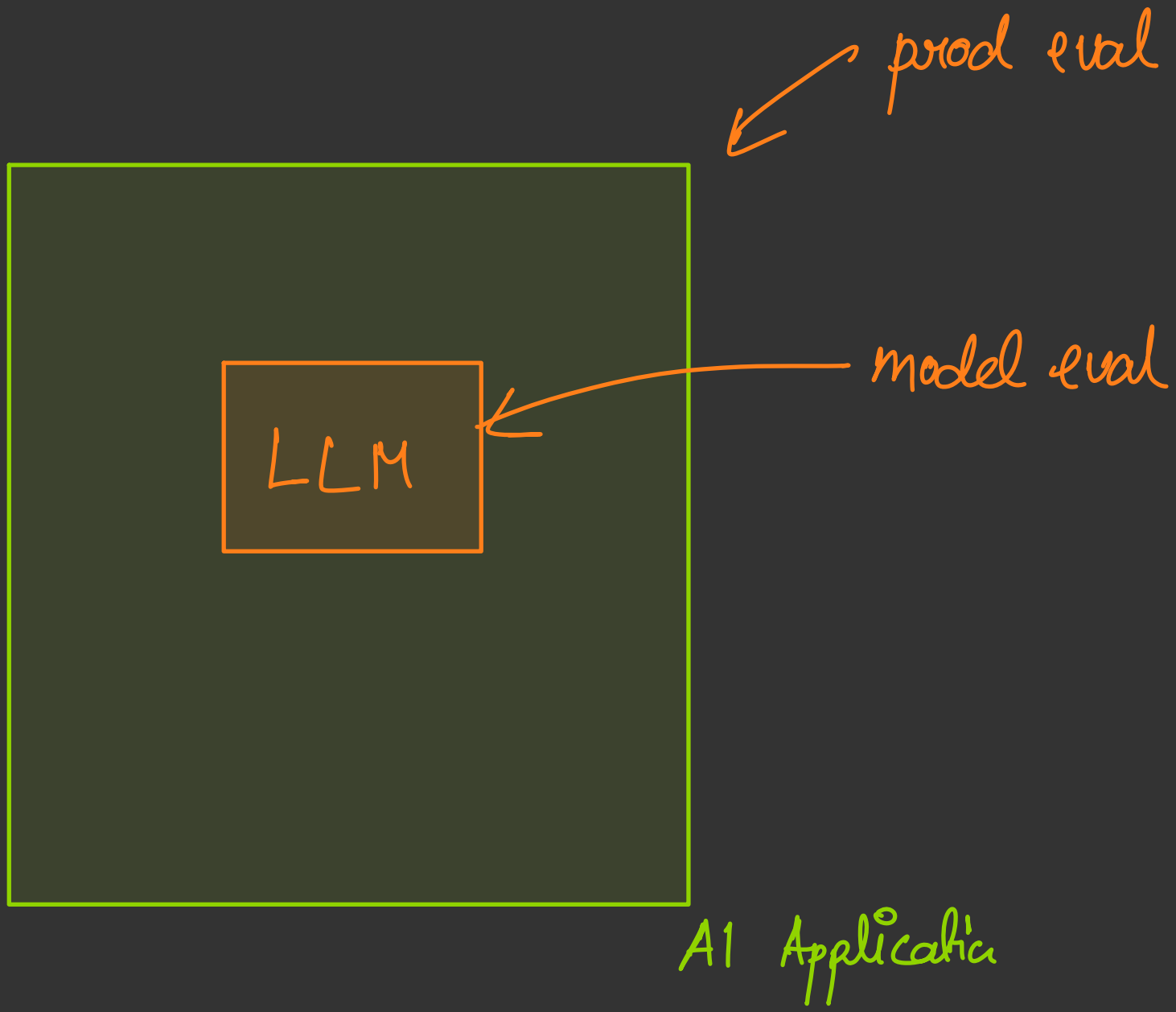
↳ define the ground truth

4) Define the evaluation prompt

↳ Write a clear & specific evaluation criteria


Q/A
Question
Ref
Answer
Candidate


Q/A
pdf
question
answer
candidate
LLM
response
Dataset
q
actual_answer


prod eval
model eval
LLM
AI Applicatio

Rouge

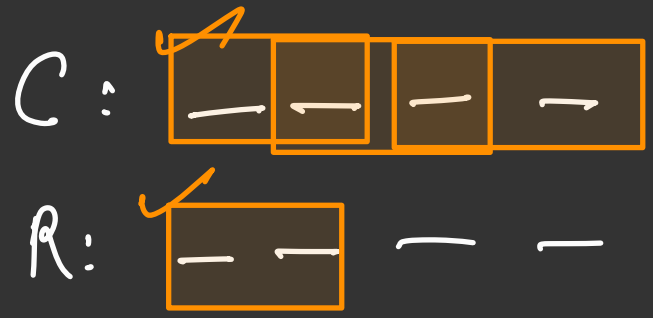
1) Rouge-1

Candidate

Reference

1 : 1 Token

2) Rouge-2

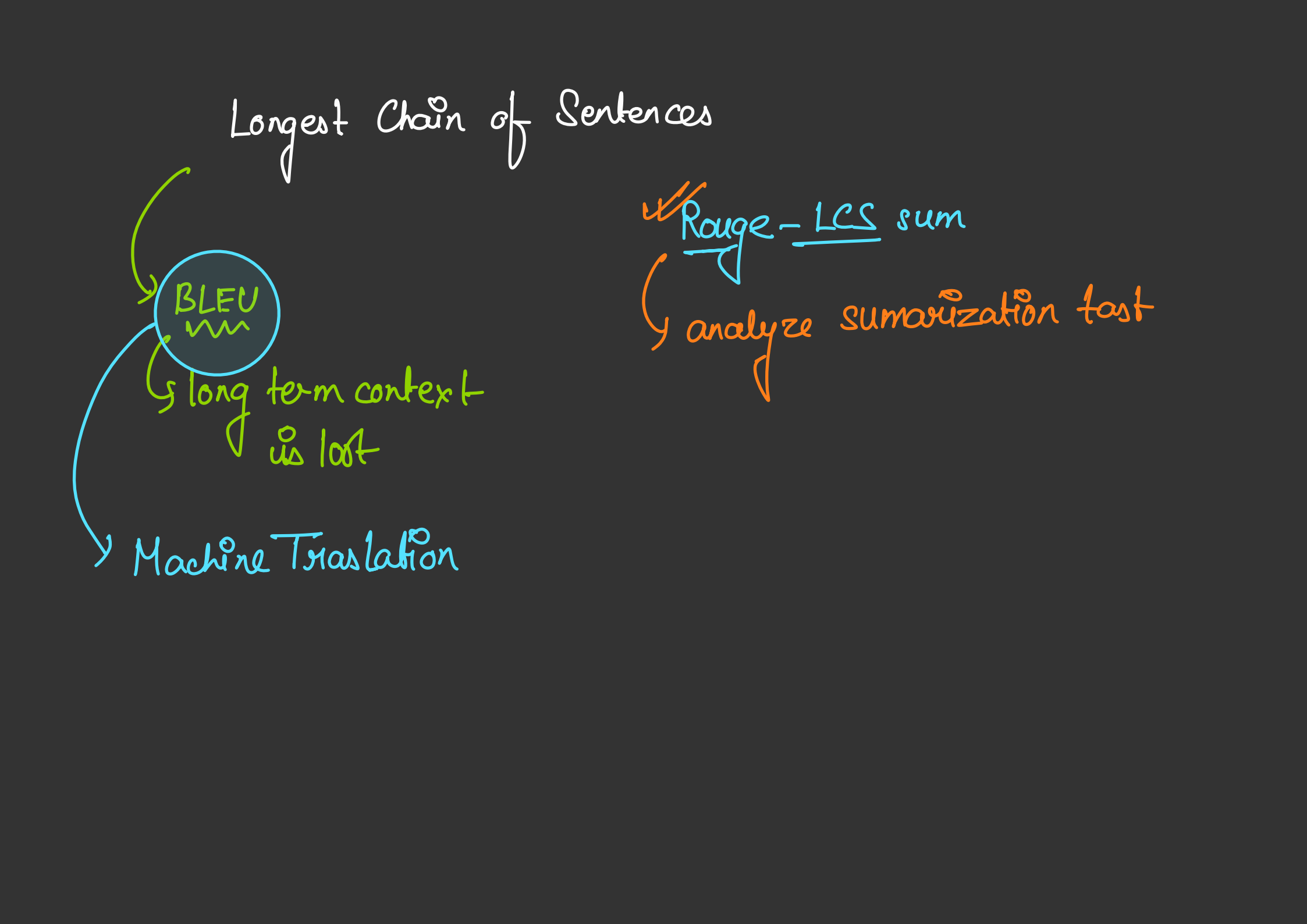


3) Rouge L

C: Hello how are you?

R: how are you?

3


Longest Chain of Sentences
BLEU
long term context is lost
Machine Translation
Rouge - LCS sum
analyze summarization task

prompt Engineering

Instruction to your LLM

1) Simple prompting → Common instruction (Nothing very specific)

a) Task definition : Solve this math problem

b) You provide context around it : 5×3 = ?

c) How the output looks like : Numbers